Detailed Evaluation of Low SNR Pulsar Data Records

Peter East

Abstract

Amateur detection of weak, low signal-to-noise ratio (SNR) pulsars with small aperture systems demands long observation times, implying extremely large data files. As a consequence, the processes necessary to prove pulsar recognition working on these large files can take a considerable time. This article offers some data processing techniques that significantly reduce the operating file sizes without losing valuable information. The result is much faster processing, tens of seconds rather than hours, allowing extraction of more and improved pulsar recognition detail than required by the usual professional validation measures [1,2]. In this paper, the methods of analyzing pulsar properties to complete validation will be described from real data collected on pulsar B0329 by a small aperture system, through to the software working on full data analysis.

Introduction

Of the various approaches to pulsar detection and recognition, a number of amateur pulsar enthusiasts are using Andrea and Giorgio Dell'Immagine's Linux software for data collection by the RTL, Airspy or Hack-RF SDRs and subsequent processing with PRESTO professional pulsar software [3].

Marcus Leech has produced his 'Stupid Simple Pulsar' software running in the GNU Radio environment for data acquisition by a wide range of SDRs; again outputting data in filterbank file format for analysis by PRESTO [4].

For the RTL SDR, Michiel Klassen has developed his 3pt tools which together with his Python GUI allows the data to be examined and presents a nice plot from which pulsar B0329 can be recognized [5].

This article describes another alternative to PRESTO, enabling improved examination of weak low SNR pulsar acquisitions. The technique involves a cut-down version of Marcus Leech's GNU Radio software for data collection followed by bespoke software for compressing and analyzing very long data recordings without losing any pulsar information. This is so that many post-processing folding algorithms can be applied to rapidly explore in detail, various pulsar recognition aspects.

The software initially implements the data compression process to speed post processing together with pulse bandwidth matched-filtering to improve the data time resolution. All pulsar validation processing is based on the standard folding algorithm. Two adaptations of conventional folding are implemented to add credence to pulsar detection by tracking pulsar pulse scintillation along the data record and these involve the techniques of cumulative and rolling-average folding. The output is 21 text files in under half a minute containing folded results that can quickly be plotted to demonstrate data conformance to known pulsar signal properties.

The article leads by describing the observation hardware for pulsar B0329+54 used to collect the data later used for software testing. The data compression and matched-filtering processes are then justified before explaining how this data is conditioned and used to produce the validation results.

The Appendix contains a summary of the pulsar analysis files plotted in MathCad.

The Hardware

The receiver system uses a pair of 2.5m Yagis (Figure 1a), tuned to the 611 MHz radio astronomy band, directly feeding a matched pair of 0.4 dB Mini-Circuits amplifiers and combined in a 3 dB in-phase splitter plus a 608 MHz to 614 MHz in-line filter box from JJ Antenna Components.

A 6 m low-loss cable feeds a receiver panel (Figure 1b) with two more low-noise amplifiers split by a second in-line filter driving a 10 MHz RF bandwidth Airspy SDR, frequency locked by a GPS disciplined oscillator (www.sdr-kits.net). The system housing comprises a large aluminium box containing a transformer fed analog power supply stabilized by a 12 volt sealed rechargeable battery, the receiver panel and a laptop PC controlling data acquisition. The box lid is closed during data recording for RFI screening .

Figure 1. a) Antenna LNAs and Combiner. **b)** Receiver Panel. **c)** System Housing and Data Collection PC.

Data Collection and Pre-Processing

Data collection is by a laptop PC using a cut-down version of Marcus Leech's 'stupid_simple_pulsar' GNU Radio software [4]. A distinct advantage of choosing GNU Radio for radio astronomy is its support for most modern SDRs - in this case the Airspy R2.

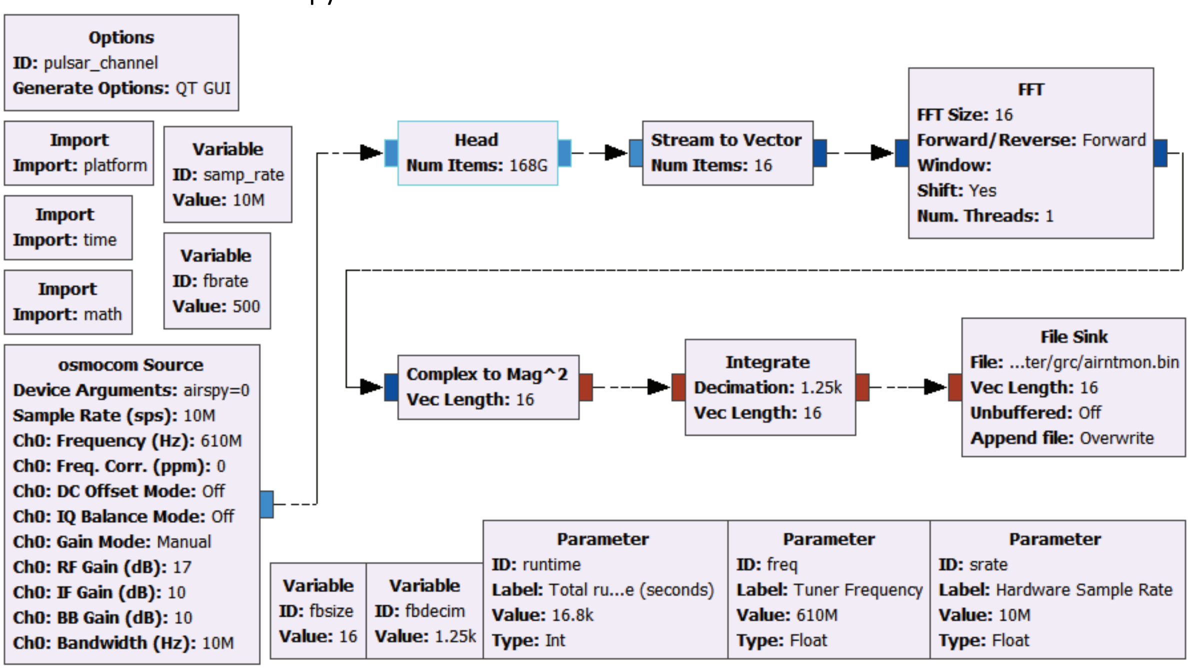

Figure 2. GNU Radio Flowchart

The 'Head' (highlighted) block in Figure 2 defines the number of complex data samples processed in the observation, determined by the observation duration in seconds specified by the 'runtime' parameter. The 'Stream to Vector' block splits the data into 16 complex streams ready for the 16-bit 'FFT block 'Size'. The FFT 16 complex outputs are then converted to 32-bit floating point magnitudes proportional to the input power in the 16 RF bands before being averaged over 1250 integration periods in the 'Integrate' block to downsize the data to 16 x 500 samples per second (S/s) rate for recording in the 'File Sink' block. The output file format is standard 32-bit binary.

To minimize amplifier effects of amplifier drift, the data recording was started an hour before the designated start time and quickly restarted at the designated start time.
With this system, even for 4+ hours intercepting pulsar B0329+54 scintillations, integrated (folded) data is only expected to produce final signal-to noise ratios of between 3:1 and 7:1.
Data was collected in the early hours of the morning of 25 October 2021; it was relatively free from RF interference and is analyzed as the example in the following sections.

Using the Airspy SDR, running at 10 MS/s and tuned to 611 MHz for 4 hr 40 mins, 168 Gs of I/Q data was sampled by the SDR and after 16-band channelization in the FFT and downsampled by running integration over 1250 samples, in this instance, the output file contained 134.4 Ms based on a 2 ms output data clock.

Pulsar Recognition Features - Weak signals
The key identifying properties of a pulsar are,

1. Regular pulse train
2. Stable averaged pulse width
3. Accurate predictable topocentric period
4. Highly stable source - known spin-down rate
5. Wide band noise source
6. Dispersed in frequency - lower frequencies follow higher frequencies
7. Predictable dispersion measure (DM), period and period change rate (P-dot) search properties
8. Random scintillating pulse amplitude with time and frequency
9. In drift-scan mode, received pulse amplitude generally follows the beam pattern

The analysis reported in this article seeks to investigate various aspects of the recorded data by using only the basic folding algorithm to check conformance with the unique properties listed above. The pulse matched-bandwidth folding algorithm has been shown to be the most efficient tool for extracting pulsar signals from noise [6].

Data Compression - Partial Folding [7]
Data compression and improving the data time resolution of the GNU Radio data with minimum information loss is attractive as it can seriously reduce the total computation time when analyzing the detailed pulsar validation characteristics to prove pulsar presence. The validation standard set by the professional pulsar detection software PRESTO is assumed the target to match. Loss-less data compression is particularly important for long observation times and weak signals as is apparent here. This can be achieved by using a synchronous, partial folding algorithm. A C-program, *syn_compress.exe* has been written to accomplish the data reduction with negligible loss of timing information and accepts the GNU Radio output files directly .

The partial folding principle of operation can be described with reference to Figure 3. The input data file can be regarded as a sequence of pulsar periods (period value as given by TEMPO). A simple method of data compression is to divide the data record into say, S equal sections and synchronously fold each section into a single period of B bins as depicted in Figure 3.

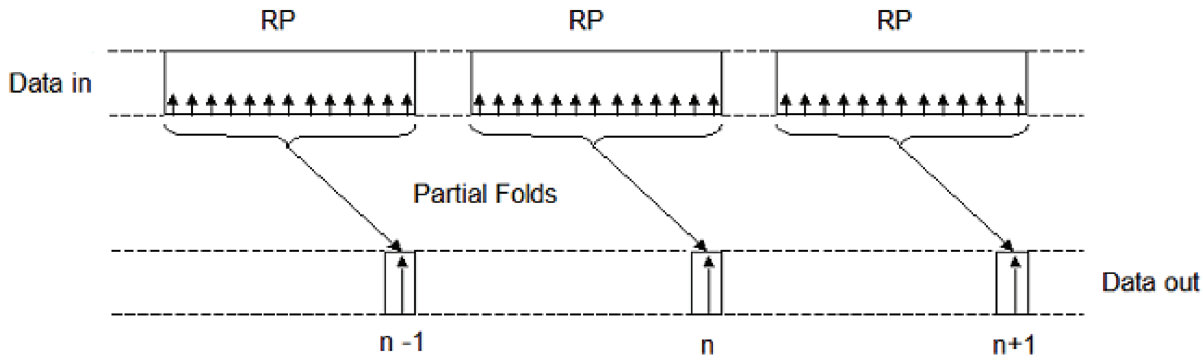

Figure 3. Partial Folding Operation

where,

P = Pulsar Topocentric period

R = Number of periods folded per Section

S = Number of Sections

n = Section Number

B = Number of fold bins = New effective compressed data period

Stacking or adding the section folds in parallel is the same as folding the whole file into one period as the standard method. On the other hand, adding the section folds in series creates another file with all the information held in the original file but now shortened by a factor, R, equal to number of pulsar periods in the input data file divided by the number of sections, S. The sensitivity to period and P-dot search parameters consequently change by the factor $1/R$. For further folding the fold period is adjusted to equal the number of bins chosen for the compressed data.

In this example, the compressed file, locked to the TEMPO derived value of the pulsar topocentric period, comprised 16-channels of 131072 samples (S = 128, B = 1024, R ~ 183) folded to approximately 0.7 ms resolution (pulsar B0329 period is approximately 714.5 ms) .

Matched-Filter Folding [8]

In standard folding, the pulsar pulse is recovered by stacking periods so the pulsar pulse adds linearly and the noise as the square root so increasing the signal-to noise ratio by the square root of the number of periods added. The summing effectively reduces the data noise bandwidth which is eventually limited in the standard algorithm by the number of fold bins. The optimum bin number, in this case, is equal to the pulsar P/W, period/pulse width ratio. But we can do better than this since we do not need to rely on averaging process to reduce the noise bandwidth, we can do this by filtering the noise to control its bandwidth to be matched to the bandwidth required to just pass the pulsar pulse shape.

Another problem with low bin numbers, apart from limited resolution, is that the pulse phase may straddle two bins affecting the perceived SNR.

The FFT can be used to band limit the data spectrum in two easy ways,

1) FFT data x modulus (FFT pulse shape) –> Inverse FFT
2) FFT data –> zero high frequencies –> Inverse FFT

This process can be carried out at any stage on the detected data, the compressed data or also on the folded data.

Figure 4 compares the standard and matched-filter bandwidth limiting methods. The green and magenta curves describe the theoretical performances for Gaussian shaped pulses of the two methods. The red and blue dots are the results on real data of modifying the fold bin number over the range 10 to 800 in 1- bin increments.

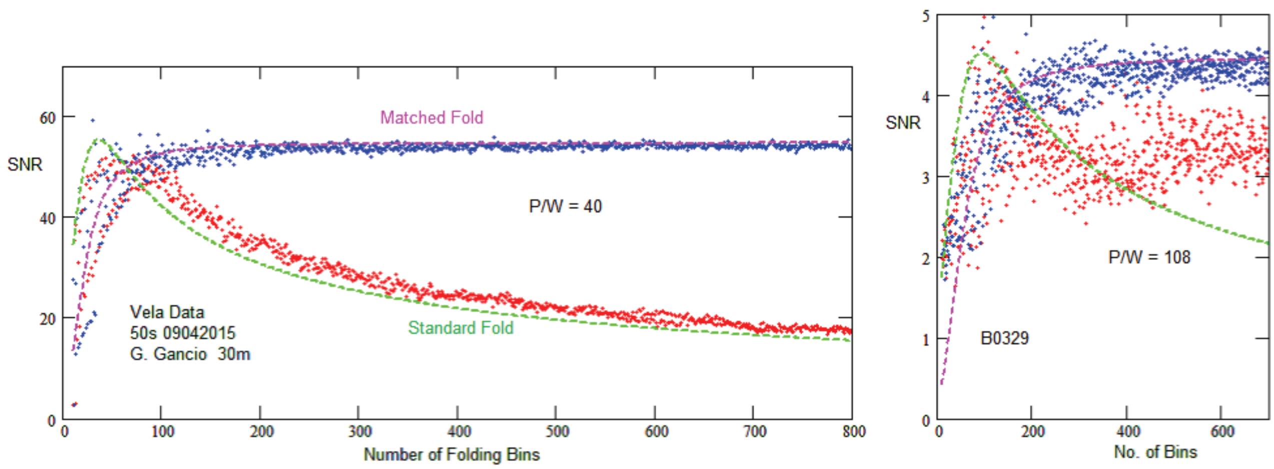

Figure 4. Comparison of Matched-filter and standard Fold Algorithms

The left hand plot uses Vela pulsar data collected by Guillermo Gancio on the Argentine 30 m radio telescope. The large antenna aperture provides a strong signal with a large SNR and the data tracks the theoretical data very well. The right-hand plot displays the performance with the weaker, much lower SNR, B0329 signal (~4.5:1 SNR) typical of small aperture amateur systems as described here. The consequences of the low SNR is a much wider measurement uncertainty. The matched filter system providing a much more confident measure whilst the standard method plateaus out on typical Gaussian noise peaks around SNR of 3:1.
The wider variation at low bin numbers is caused by the pulsar pulse straddling bins.

Data Conditioning

The Figure 5 processing chart summarizes the data processing operation sequence for extracting pulsar characteristics generally used to validate true pulsar detection. The chart shows the recommended data reduction paths to usefully speed up the computation times, particularly for the more computationally intensive 2-D period/P-dot search plot.

Data Compression and Matched Filtering

In this example, the GNU Radio detected data file comprised 16 frequency channels each of 8.4MS of video data at 2ms clock rate. The compression and matched filtering functions reduced this to 16 channels of 128 sections, each synchronously partially folded into 1024 bins. This process loses no timing information as it can be shown that folding/summing the 128 sections produces an identical result as if the whole data were folded prior to the compression process. There is no observable difference if the GNU Radio data were produced at a 1 ms or lower clock rate, even though the final folded result now indicate, sampling at approximately 0.7 ms intervals.
In the data compression process, DC removal of each section prior to re-composition was essential to ensure that any low frequency drift over the observation period did not corrupt the resulting data with DC steps between sections.

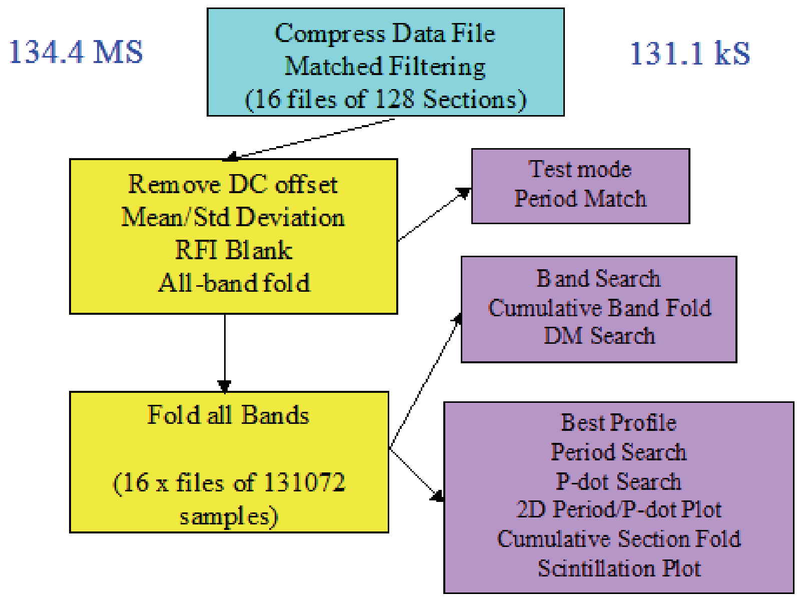

Figure 5. Validation Processing Chart

RFI Blanking

The 16 frequency band files each comprise 131072 samples. In this example 16-bands was chosen so that if a narrow band interfering signal was present in the data, that channel could be blanked without losing too much overall bandwidth. Similarly if one or more sections were obviously corrupt, these can also be blanked. The blanking method employed uses two text files identifying the channel band and/or sections to be blanked.

The number of folding bins was set at 1024 to define the final timing resolution (for B0329) to be close to 0.7 ms. Similarly 128 sections was chosen so that if one or more of these were affected by severe RFI then blanking a few of these may be an acceptable data loss.

In this case, night time data collection ensured that locally generated RFI was minimized.

Extracting Pulsar Characteristics

The final stage, once the data has been suitably conditioned is to extract the target pulsar's detailed characteristics by various folding arrangements. Initially, all the compressed band data are folded both individually, then all together with the bands combined.

The software has a quick reaction test mode that just carries out period and P-dot searches with data presented in the command terminal so that minor adjustments can be made to the TEMPO defined topocentric period.

Once period adjustment has been set, the full analysis, producing 21 specific space-separated text files that can be examined using Excel, Notepad, MathCad MatLab or Python to investigate pulsar parameters in detail. These are demonstrated in the following sections.

Software Tool

The post processing software, *syn_compress.exe* runs in a 64-bit WINDOWS command terminal (*cmd.exe*) .

The command calling terms are of the form;-

syn_compress <infile> <No. FFT bins><data clock (ms))><period (ms)> <No. sections><No. bins><pulse width><DM><ppm adjust><Test T><RF bandwidth (Mhz)><RF Center (MHz)><Roll average No.>

Example:

syn_compress.exe data.bin 16 2 714.492092 128 1024 6.5 26.7 0.1 04 10 611 32

where,

data.bin is the input data file generated by GNU Radio as a 32-bit binary file of detected data samples
16 is the number of FFT channels/bands.
2 is the Gnu Radio set data clock in ms.
714.492092 is the TEMPO topocentric period for the observation site and recording UTC
128 and *1024* are the numbers of compressed file sections and fold bins - both binary numbers.
6.5 is the target pulsar pulse width.
26.7 is the target pulsar dispersion measure
0.1 is the period correction offset in ppm.
04/T4, *0* or *T* , the first character, specifies normal or test mode. *4* is the period range divisor.
10 is the SDR RF bandwidth in MHz
611 is the RF center frequency
32 is the rolling average section range.

The Command Terminal Response reports the key parameter data:-
P:\grc>syn_compress.exe airntmon.bin 16 2 714.492092 128 1024 6.5 26.7 0 04 10 611 32

```
Mode = 0  Period Range Divisor = 2
Pulsar Period = 714.492092 ms
Data clock= 2.00 ms
Pulsar Pulse Width = 6.50 ms
Pulsar Dispersion Measure = 26.70
Rolling Average Number = 8
Input file bytes =537600000
RF Centre = 611.0 MHz,  RF Bandwidth = 10.0 MHz
DM Band Delay = 9.716ms, No. Delay bins = 13.924
Blanked Bands: 0
Blanked Sections: 0
No. Data Samples = 134400000
No: FFTs = 16
Input Data per FFT channel = 8400000
No. of Output fold blocks  128 ;   bins = 1024
No. of Output samples = 131072
Working file duration = 16800 secs
Number of pulsar periods = 23513
ppm adjustment = 0.00
Max Pulse ppm drift = 0.00 ms
Compression Ratio = 183.70
Period Search Range = -12.50 ppm to 12.50 ppm
P-dot Search Range = -10.63 pp10e10 to 10.63 pp10e10
```

Blank data files:

The software expects two blanking files to be available. These are, *Blanks.txt* and *Blankf.txt*
These are standard space separated number files, identifying section and band numbers to be blanked for RFI or bad data reasons. They can be empty files where all data is to be processed. Data to occupy the files can be obtained from the output file plot analysis.

Processed Output Data Files and content description:-

1. Raw compressed channelized data file: rawdat.txt
2. Compressed filtered channelized data file: outdat.txt
3. Compressed filtered combined bands data file: allbands.txt
4. Pulse profile: profile.txt
5. Period search: periodS.txt
6. P-dot search: pdotS.txt
7. 2-D Period/P-dot peak SNR: ppd2d.txt
8. DM search: dmSearch.txt
9. DM search target blanked: dmSrchns.txt
10. DM peak best fold data: dmprf.txt
11. Band Peak SNR: bandS.txt
12. Folded bands: bandat.txt
13. Cumulative band Peak SNR: cumbands.txt
14. Cumulative folded bands: bndcum.txt
15. Section peak SNR: secsnr.txt
16. Section folded data: puldat.txt
17. Cumulative section peak SNR: cumsec.txt
18. Cumulative folded section data: foldat.txt.
19. Rolling average section peak SNR: secavsnr.tx
20. Rolling average section folds: secavfol.txt
21. Rolling section block peak SNR: secavrol.txt

These files are used to construct the plots in the following Results and Appendix sections. Running the software tool on a modern 64-bit computer, the run time for the test mode is about 10 seconds and for the full analysis mode is about 25 seconds. Output data files used for the analysis figures following are identified in numbered brackets at the end of the figure titles.

Test Mode - Period Correction

Mode = T Period Range Divisor = 4

With any amateur potential pulsar data collection, the pulsar identity and position is known and the data collection set up accordingly. Knowing the time, one can use TEMPO to predict the likely observed topocentric period and this is the usual fold period used in the subsequent data search. Due to receiver clock limitations, the measured period may be slightly different to the predicted figure requiring a search sequence.

ppm = -0.75	SNR = 3.22	bin = 401	pdot = -0.64	SNR = 3.76	bin = 61
ppm = -0.50	SNR = 3.51	bin = 643	pdot = -0.43	SNR = 3.34	bin = 60
ppm = -0.25	SNR = 4.43	bin = 640	pdot = -0.21	SNR = 4.40	bin = 639
ppm = 0.00	**SNR = 4.85**	**bin = 637**	**pdot = 0.00**	**SNR = 4.85**	**bin = 637**
ppm = 0.25	SNR = 4.74	bin = 634	pdot = 0.21	SNR = 4.69	bin = 635
ppm = 0.50	SNR = 4.38	bin = 632	pdot = 0.43	SNR = 4.21	bin = 634

Table 1 Period Adjust Error Indicator

The software designed here, therefore has a fast testing mode that lists a period search range in the command terminal that may be inspected to determine any possible error which can then be added in the software command line to correct the deviation and revert to full analysis mode. In Table 1, the maximum SNR is indicated at zero period ppm error so needs no adjustment. Note that the P-dot error is also zero, implying negligible period rate drift over the observation time confirming a very stable source.

Results

Full Data Folding

Figure 6 plots the best folded result (red) and the peak DM search folded result (blue). The unexpected difference is because the best result is obtained by a simple method of better utilizing the group of sections folded and is discussed later.

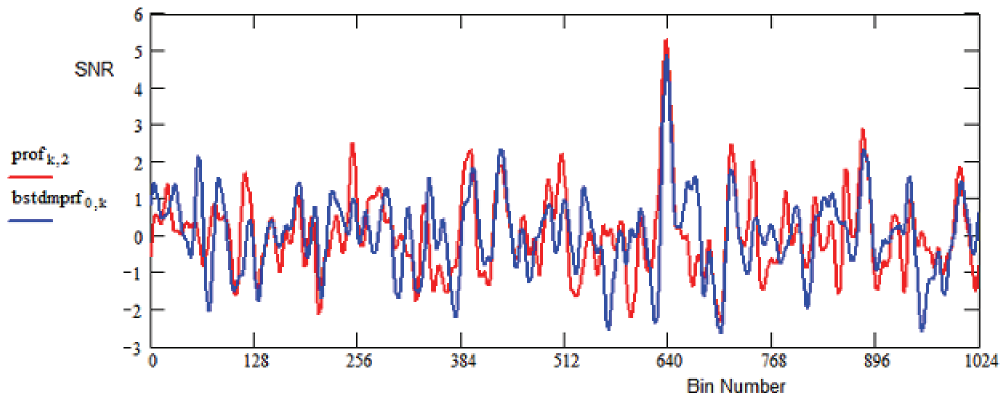

Figure 6. Matched Topocentric Period Folded Data (4,10)

Period/P-dot Search

Period and period rate variations about the expected values confirm both pulsar period accuracy and negligible period variation over the observation time.

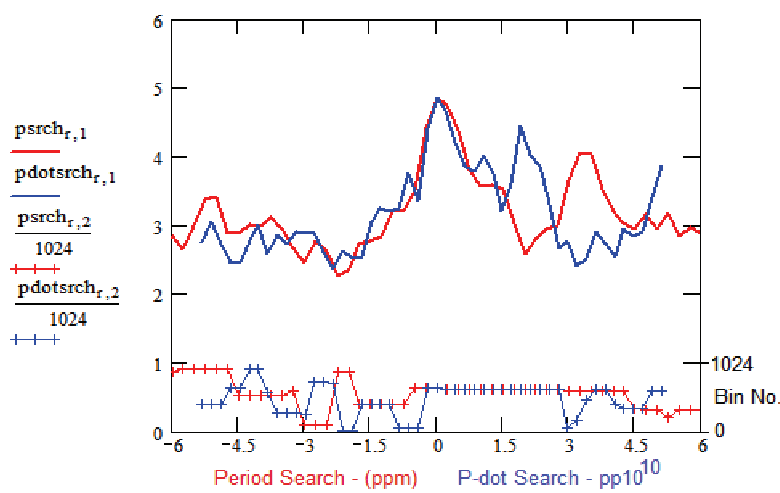

Figure 7. Period/P-dot Search (5,6)

The period/P-dot plot above also includes the bin number at which the moving peaks occur. With both these changes, the target pulsar peak varies at a predictable rate, determined by the extent in the record that it is observed. With low SNR values as recorded here, local noise peaks tend to take over affecting the data quality.

With both parameters peaking at zero offsets, this test proves exact match to the pulsar expected topocentric period and confirms the stability with no period drift within the 4+ hour observation time.

2D Period/P-dot Plot
Figure 9 aims to explain the operation and characteristics expected from the typical 2-D period/P-dot search plot of Figure 8. The test examines the peak response of data folded over a range of period values around the period topocentric value and over a positive and negative range of P-dot values to explain the 'ridge' observed. The theoretical ridge slope is given by $pd/p = -2/N$, where *pd* and *p* represent the P-dot and period offset values and *N* is the total number of pulsar periods in the observation record. For Figure 8, the scales have been adjusted to make the correct ridge slope occur at -45 degrees.

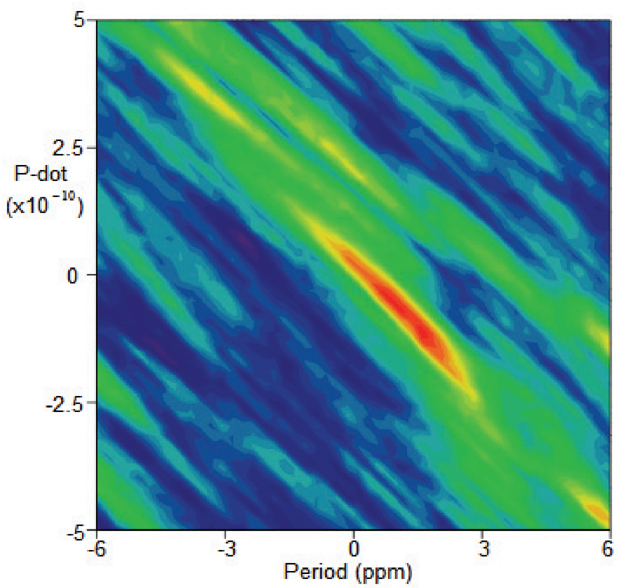

Figure 8. 2-D Period/P-dot Search (7)

A clear central peak at coordinates 0,0 and sloping extended ridge is evident confirming period match and nominally zero spin-down.

An explanation for the ridge effect follows. Addressing Figure 9, the central green horizontal line with notional period intervals indicated corresponds to the test period value matching the pulsar topocentric period together with a zero P-dot value. The result is the horizontal green maximum fold response to the right.
The magenta line represents a period positive offset, still with zero P-dot, which tends to spread the (magenta) folded result to the right and lowering its amplitude as shown.
The red P-dot line starts with a test period matching topocentric figure together with a P-dot value which progressively reduces the folding test period resulting in the red folded response to the right. Whereas the period offset response is symmetrical, the P-dot starting at zero and working negative, shortens the folding test period, moving it from a matched condition to unmatched exhibiting an asymmetrical (red) response peaking closer to the matched position.
Finally, the blue P-dot line on the left starts with a positive period offset coupled with a P-dot value which progressively shortens the period over the observed periods such that the final period matches the original topocentric period resulting in the much increased blue folded response the right.

The black ellipse indicates the range within which combined opposite polarity period and P-dot offset values can give rise to the stronger fold (blue) peak.

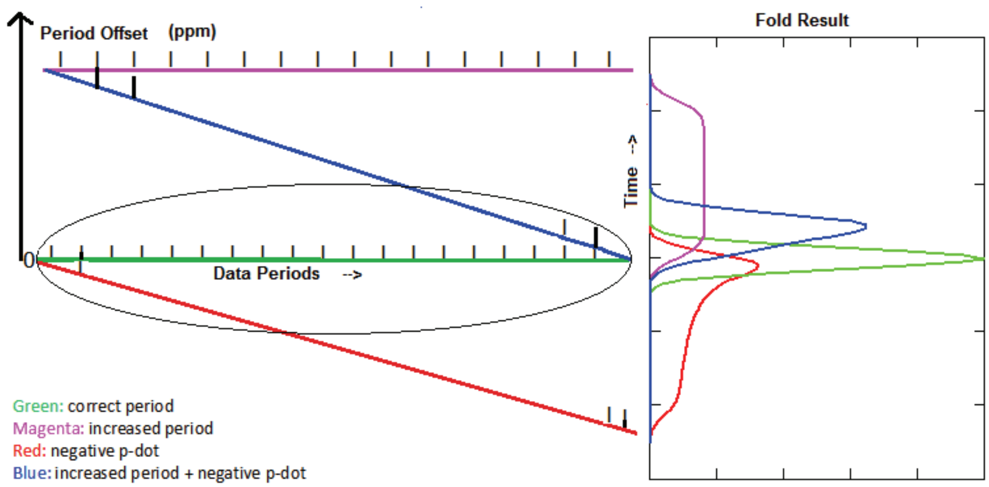

Figure 9. 2-D Period/P-dot Operating Characteristics

This compensating property of opposite polarity period and P-dot offsets is responsible for the extended diagonal ridge observed through the center of the 2-D period/P-dot search plot. Figure 9 can explain the extended ridge observed in the 2-D plot by noting that for a any given positive period offset (magenta) a very wide range of negative P-dot slope values (blue) can produce a period range closely matching the data period (within the black ellipse). The closer the initial period offset, the larger the fold amplitude contributing to the ridge.

Efficient and timely computation of the 2-D Period/P-dot search plot is the main reason for wanting to compress the data. The separate period search and P-dot search plots are relatively easy to produce, typically requiring less than a hundred iterations to plot the relevant sub-plots in Figure 7. To calculate the 2-D plot requires some 2500 iterations. With multi-mega-sample files the computation time can be very long but by compressing the data to 131072 samples or so the computation time becomes more acceptable.

Dispersion Search

Data de-dispersion involves separating the received signal bandwidth into a number of sub-bands and delaying the higher sub-bands before video or digital summing to compensate for the interstellar delays.

For dispersion measure search, the band delays are varied linearly over a range about the band center to accomplish both positive and negative dispersions; the combined data SNR is then calculated to produce a plot similar to the red crossed plot in Figure 10.

A simple check that it is indeed the candidate pulse that is responsible for the dispersion search peak around DM = 26.7 is to blank the pulse response around the candidate (in this case, bin 637) and repeat the search routine (magenta curve). It is seen that subtracting the two measurements now produces the blue curve which is identified as the DM contribution of the nulled candidate peak. This gives a much clearer indication that pulse at bin 637 peaking around DM = 26.7 and confirms the dispersion measure expected for pulsar B0329.

The horizontal crossed red plot reports the bin number of the peak response (scale to the right), noting that it is indeed constant and correctly reporting the target bin number. The figure remains constant as the band positive and negative band delays are maintained relative to the total band center.

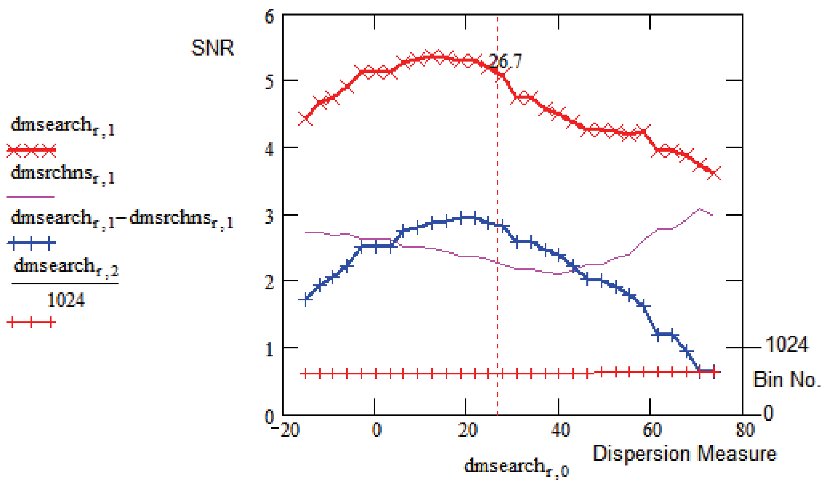

Figure 10. Improving Dispersion Search Discrimination for low SNRs (8,9)

Once the band delays that produce the maximum pulsar amplitude has been identified, the sub-band files are delayed as required and the files added to produce a single file to support the remaining validation processes.

Cumulative Frequency Waterfall

Waterfall plots are usually constructed by dividing the data into a number of equal sections, processing these as required, then displaying the resulting plots sequentially. With cumulative waterfalls, the data is again divided into the same equal sections, but now displaying the processed results of the sum of all previous sections. This improves visibility of weak information that may be lost if noise when splitting the data in many sections.

Here, the sixteen frequency band files are cumulatively folded and the peak surface plotted to produce Figure 11.

To start cumulative folding band zero is first folded. then bands 0 and 1 are summed and the sum folded.

This process continues until all bands are summed and folded. It can be seen from Figure 11 that once 5 bands are summed and folded that the target pulsar (red line) starts to dominate.

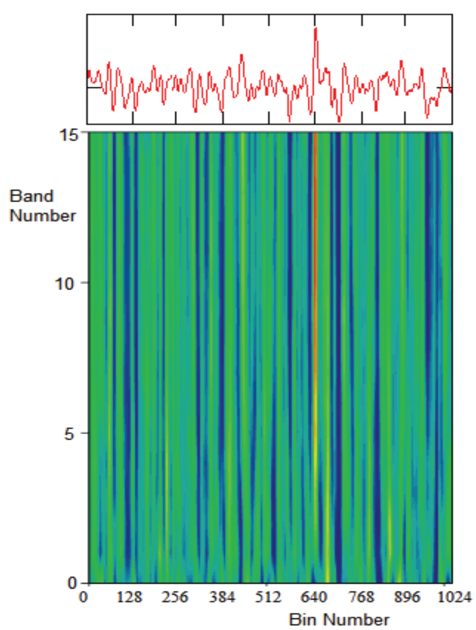

Figure 11. Frequency Waterfall (4,13)

With the conventional waterfall method, where only individual bands are folded for display which for weak pulsar signals does not give a useful presentation. This process continues until all bands are summed and folded. This plot confirms the broad band nature of the target.

Band Response Analysis

Figure 12 presentation provides more evidence of the target wide frequency coverage. The red plot indicates the peak SNR observed as the bands are cumulatively summed and folded. After folding bands 0,1, 2 and 3 which seem to track noise peaks, it is seen there is an increasing trend in the summed and folded SNR, leveling off after the folded sum up to band 10. The green dashed plots track the individual band signal + noise fold peaks, whilst the solid red plots and dashed red lines present the cumulative and individual band SNRs of the target pulsar.

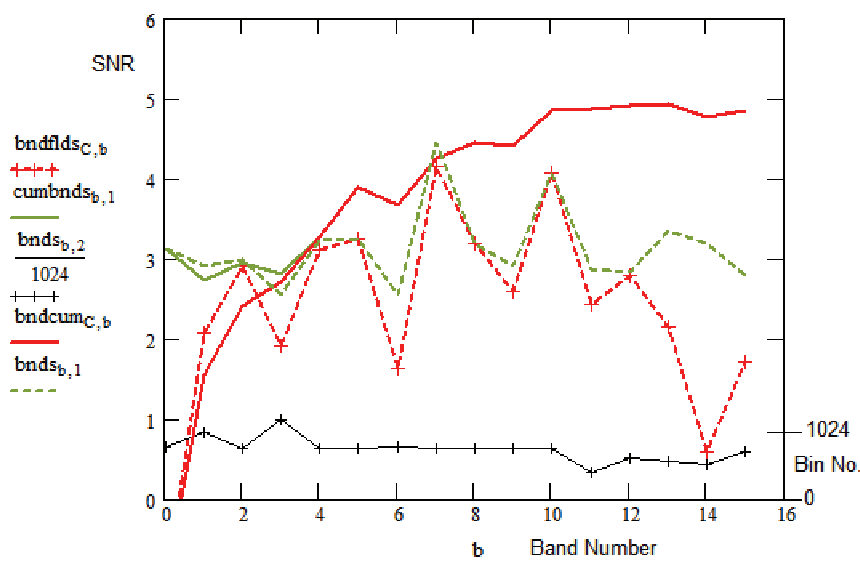

Figure 12. Cumulative Frequency Band peak SNR (red) (11,12,13,14)

Finally, the thin black crossed point line describes the individual band peak bin value (= 637, scale on right-hand axis). This shows that the target peak dominates between bands 4 and 10. With a fully integrated SNR of 5:1, it is calculated that any individual band SNR exceeding $5/\sqrt{16} = 1.25:1$ can contribute to the final figure - in this case, all bands except 0 and 14 contain useful components. It can be concluded that blanking bands 0 and 14 should lead to an increased final result.

The frequency band analysis clearly demonstrates that the target pulse source is broad band, the folded band peaks occur at the target pulsar characteristic bin and the varying levels in each band is a sign of the pulsar frequency scintillation property.

Cumulative *Time Waterfall*

The red line under the peak bin 637 in Figure 13 indicates that from about section 50. the integrated bin position 637 is becoming dominant resulting in the profile pattern in Figure 6.

With low SNR data, the cumulative waterfall representation is clearer than the conventional section SNR form, where the section pulse SNR is prone to drop below noise by the factor $\sqrt{(128)}$.

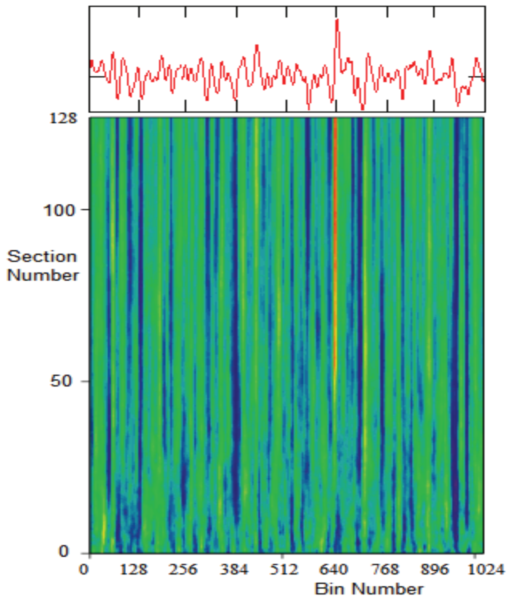

Figure 13. Time/Section Cumulative Waterfall (4,18)

The vertical green noise streaks appear typical of persistent natural folded noise peaks that once established seem to maintain a fairly flat profile.
Both waterfall results confirm the presence of a regular pulse train.

Cumulative SNR
To produce the cumulative SNR plot, the combined de-dispersed file is split into 128 sections, each section is folded and added to all previous sections. The accumulating summed folds are then checked for a maximum and this is plotted against section number in Figure 14 to produce the green curve (peak of noise or target pulse). Also plotted in red is the target pulsar at bin 637. This shows that the candidate takes over the integrated peak by about section number 50. Studying the candidate curve, there are several regions of sharp rises which can be attributed to strong scintillation. There are some negative sloping regions where the target signal is weak or subsumed in background noise. The rises are associated with 'strong' section SNRs and the falls associated with weak or negative section SNRs. The green crosses line represents the characteristic bin number of the accumulated peak and again shows the target bin number remains constant at 637 (the target pulsar bin peak) from about section 48.

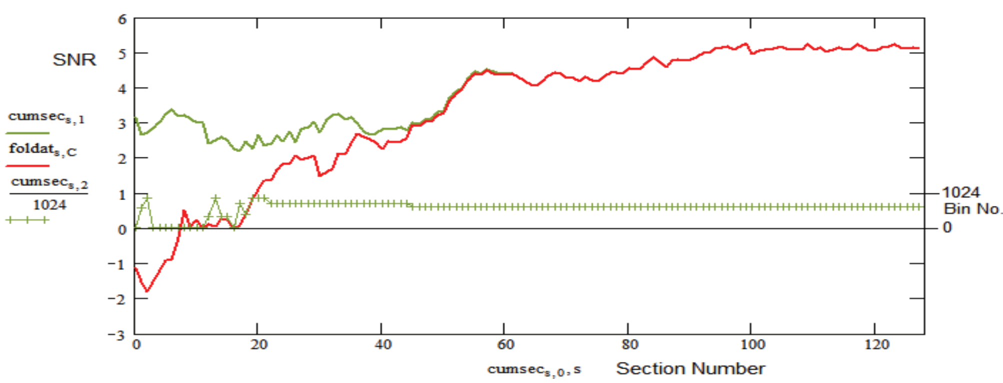

Figure 14. Cumulative SNR - noise ,green; bin 637,red (17,18)

To clarify the characteristics of the background noise, Figure 15 plots the apparent SNR of all noise components with the target pulsar bin region blanked. From this it appears that the accumulated response remains noise-like and does not exhibit the growing trend of a true semi-persistent pulse train. (bins 0 to 630, blue; 643 to 1023, green)

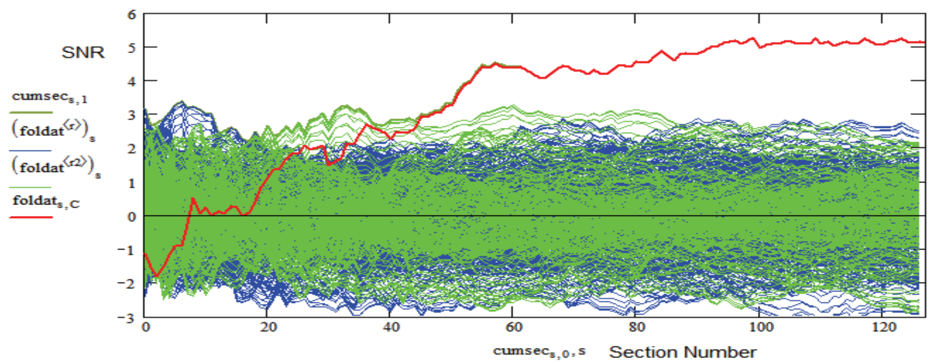

Figure 15. Cumulative SNR Plot for Target/Noise Comparison (17,18)

Scintillation

A particular problem in intercepting Pulsar B0329 is its feature of short and long-term scintillation caused by non-linearities and temporal changes in the interstellar medium.

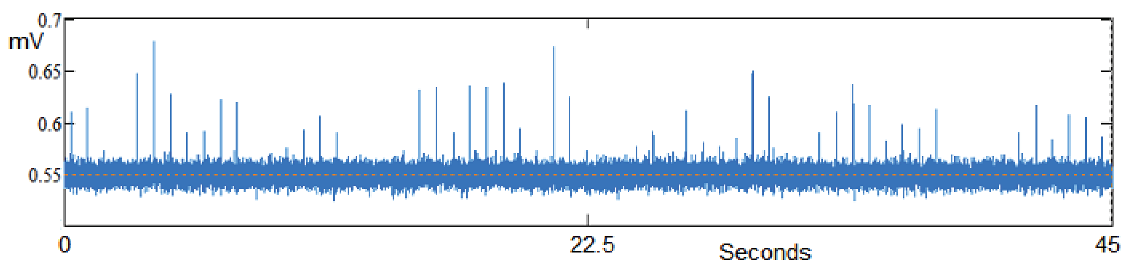

Figure 16a. Pulse-by Pulse B0329 Recording - 46m antenna

This causes long and short timescale variations in amplitude and is also variable in frequency. The blue plot in Figure 16a is a 45 second pulse-by-pulse recording taken with the 46 m dish at the Stanford Research Institute (by R. Ferranti) and clearly shows the short-term pulse-by-pulse scintillation.

Figure 16b plot is the present 1.6 m^2 aperture recording and represents the full compressed 4.67 hour, 128 section recording. Each section comprises 183 synchronously folded pulsar periods; even so, no integrated pulses are visible.

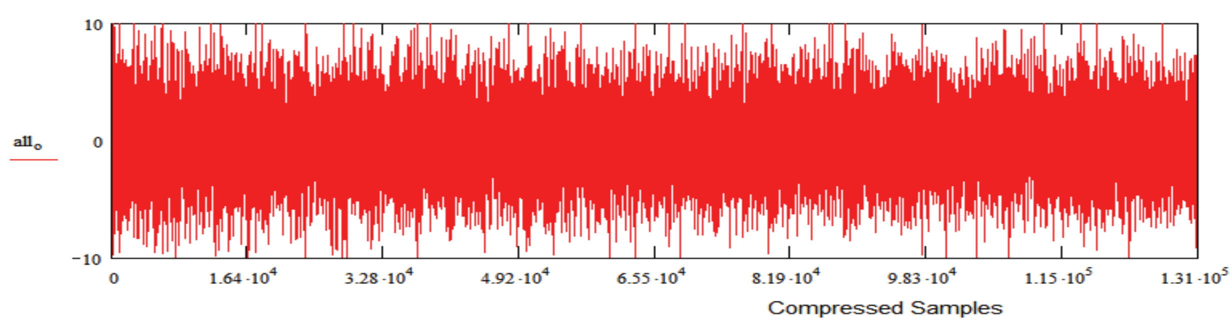

Figure 16b. Compressed Video Record - 1.5 m^2 antenna (3)

However, since the analysis has identified the bin number corresponding to the target pulsar relative position, the pulsar bin 637 contents can be selected for display as shown in Figure 16c.

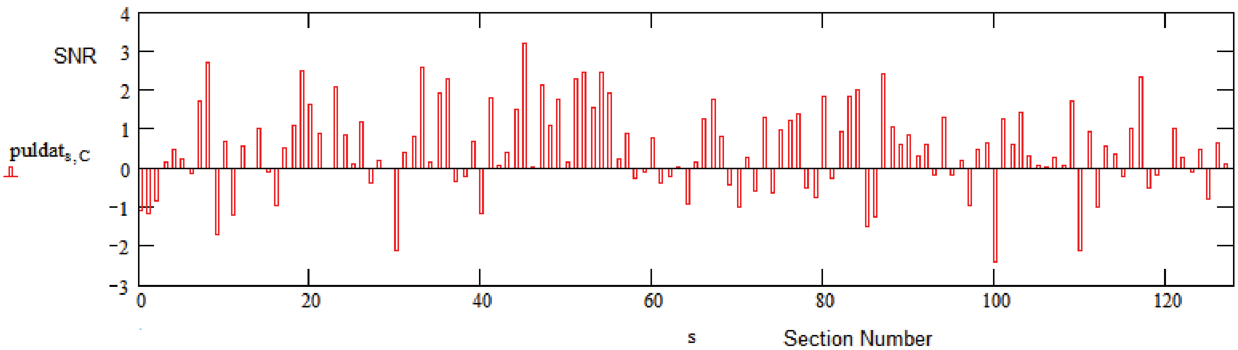

Figure 16c. 1.5 m^2 antenna Extracted Target Bin (3)

This shows that although the pulse amplitude is comparable to the noise level, there is a majority of positive, variable amplitude indications and groupings indicating both scintillation and confirming the pulsar signal presence. Caution is needed when interpreting individual responses due to the ambient noise level which is only suppressed by averaging in folding.

Data Searching

Another method of demonstrating the presence of scintillation is by constructing a rolling-block average along the data record. In Figure 17a, the red curve is again the cumulative sum of previous section folds. The blue curve is now the rolling average of 14 section folds of the target; the value displayed at the average mid-point.

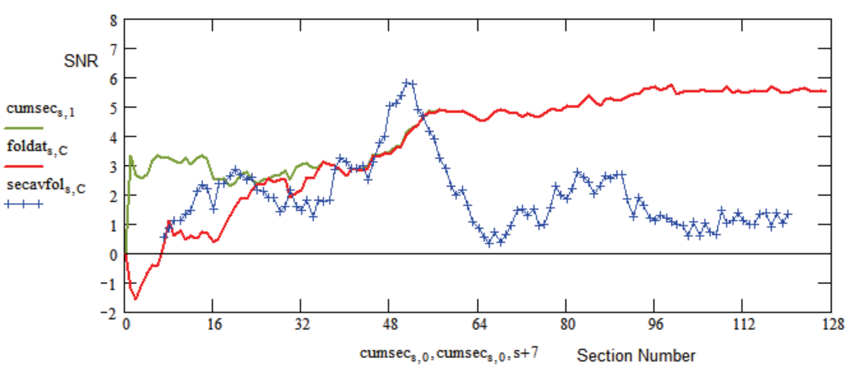

Figure 17a. 14-section Rolling Average Indicating Scintillation (17,18,20)

This rolling average technique is sufficient to better indicate the data section regions where major contributions to the final folded result reside. In Figure 17a, the dominant region appears around section 48, coinciding with the sharp slope increase of the cumulative sum. There are also strong regions in sections before and after section 48 - another indication of scintillation. Another point to note is that around section 50, the integrated SNR from the 14 sections averaged is greater than the final full data integrated fold. 14 sections corresponds to just 31 minutes of observation time demonstrating how positive scintillation can help as well as often hindering successful observations.

As well as indicating scintillation, strong points the general amplitude trend can reflect the antenna beam shape when observations are carried out in the drift scan mode. The beam shape is evident for this example when increasing the rolling block average to 32 sections as shown in Figure 17b. What is interesting is that the peak SNR indicated by this 32-section rolling average significantly exceeds the full data final SNR result and accounts for the superior SNR witnessed in the original profile of Figure 6.

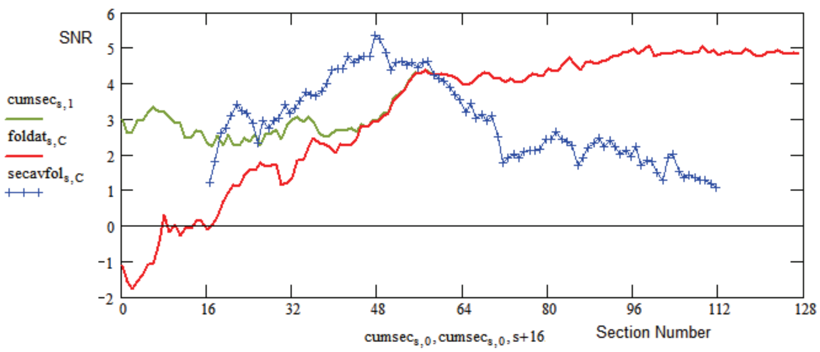

Figure 17b. 32-section Rolling Average Indicating Antenna Beamshape (17,18,20)

Another interesting comparison is offered in Figure 18. Here, the red curve is the section peak SNR (signal+noise) with an 8-section rolling average compared to the blue curve which plots the target pulsar 8-section rolling average SNR. Around section 50 the target pulsar response predominates, whereas the noise peak SNR dominates over most of the range fluctuating around 3:1 SNR level.

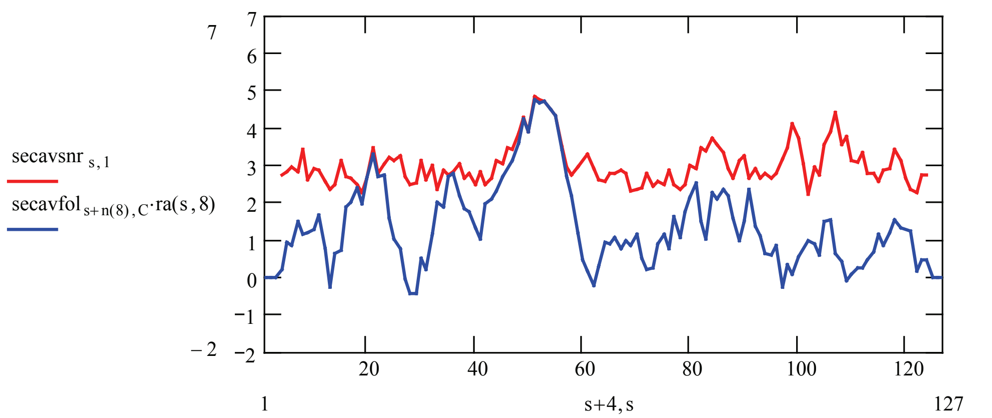

Figure 18. 8-section Rolling Average Pulsar/Noise Comparison (19,20)

Best Folding Range

A comparison of these rolling average plots hints that there is an optimum folding data range that ensures the best fold result. The red curve in Figure 19 plots the peak folded result as a function of the number of sections averaged as the whole data is scanned rolling from a single section to all sections averaged. In this example, it is noted that the curve peaks at N = 96 sections averaged but that over most section values the result is fairly high above 80 section rolling average but that it trails off above 120 sections averaged.

The blue curve is the N = 96 rolling average section scan showing that optimum scan center is at section 51. Note the rolling average scan range of 96 sections is only possible centered on bins 48 to bin 79.

The output files 20 and 21 compute the data for these plots.

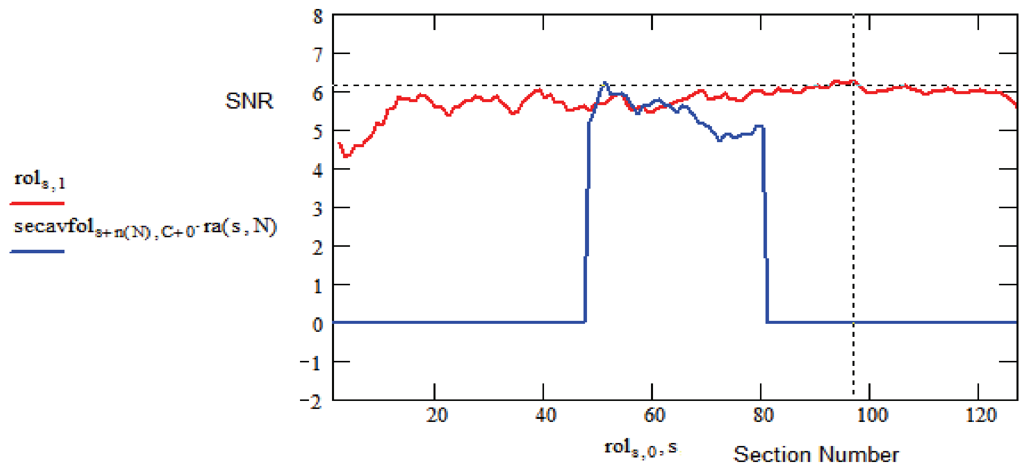

Figure 19 Rolling Average Optimum Range Selection (20,21)

It is interesting to note, that while folding the raw data at the correct topocentric period checked at the period/P-dot test phase was measured as 4.85:1, after optimum de-dispersing, this rose to 5.11:1, this best rolling average adjustment increased the measured value to 6.25:1. Further improvements could be obtained by blanking low SNR bands 0 and 14 (see Figure 12). In addition blanking sections 30,100 and 112, the three largest target negative SNR sections (see Figure 16c), raises the final SNR to 7.02:1 (Figure 20).
Blanking all negative SNR sections (see Figure 16c) appears even more beneficial, but without inspecting the data directly, because of contributions being comparable to the pulse level, this less easy to justify.

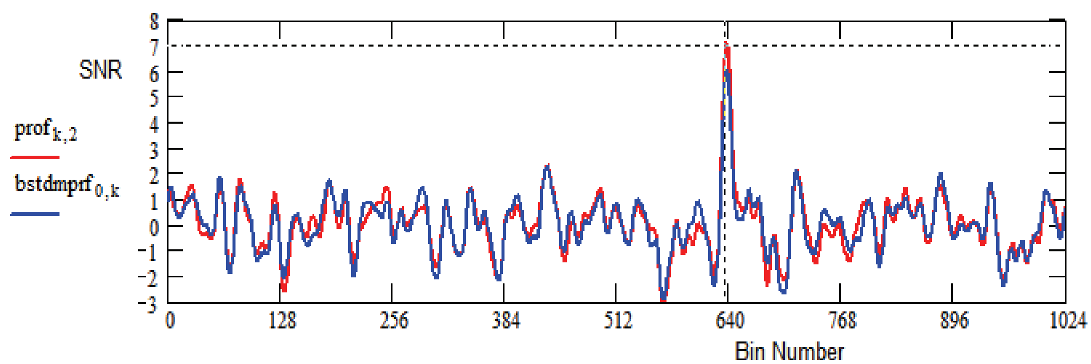

Figure 20. Improved SNR with Bands 0,14 and Sections 30,100 and 114 Blanked and Optimum Rolling Average

Conclusions

An alternative software solution for low signal-to-noise ratio pulsar data analysis, compatible with GNU Radio controlled SDR data collection and pre-processing has been described. The post-processing technique involves data compression for fast analysis of long observations whilst maintaining accurate timing. The software produces a large number of folded data files describing a wide variety of information needed for accurate measurement and validation of the various pulsar identifying parameters - some in detail not previously recognized. The advantages of the data compression method include minimal information loss together with a significant reduction in computing time. In the example described, 16800 seconds of recording time in 10MHz RF bandwidth was post-processed, producing 21 folding analysis files in 25 seconds, highlighting different pulsar validation properties. MathCad was used to plot the data files, but most are compatible with Excel. A Python GUI is to be investigated.

The cumulative SNR and rolling section averaging approaches applied to both synchronized compressed sections and frequency bands has proved very fast and useful for identifying very weak pulsar pulse trains in recorded data.
The Appendix contains MathCad plots of the relevant data file information supporting pulsar validation.

References

[1] PRESTO Home, https://www.cv.nrao.edu/~sransom/presto/

[2] SM Ransom. Searching for Pulsars with PRESTO https://www.cv.nrao.edu/~sransom/PRESTO_search_tutorial.pdf

[3] G Dell'Immagine, A Dell'Immagine. Linux pulsar software for recording and analysis. https://github.com/gio54321/pulsar-distro-guide

[4] M Leech. https://www.ccera.ca/papers/memo-12-a-pulsar-observing-capability-at-ccera/

[5] M Klaassen. http://parac.eu/projectmk17b.htm

[6] V Morello, et al. Optimal periodicity searching: Revisiting the Fast Folding Algorithm for large-scale pulsar surveys. arXiv:2004.03701v2 [astro-ph.IM], 3 Aug 2020.

[7] PW East. An Analytical Method of Recognizing Pulsars at Moderate SNR., Journal of the Society of Amateur Radio Astronomers. November-December 2018.

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PW East April 2022

Peter East, *pe@y1pwe.co.uk* is retired from a Defense Electronics career in radar and electronic warfare system design. He has authored a book on Microwave System Design Tools, is a member of the British Astronomical Association since the early '70s and joined SARA in 2013. He has had a lifelong interest in radio astronomy; presently active in amateur detection of pulsars using SDRs, and researching low SNR pulsar recognition and analysis. He has recently written another book, 'Galactic Hydrogen and Pulsars - an Amateurs Radio Astronomy' describing his work in Radio Astronomy.
He maintains an active RA website at *http://www.y1pwe.co.uk*

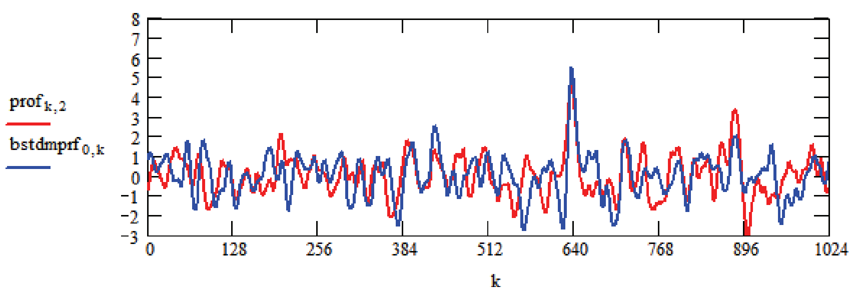
Appendix. MathCad Plots

pulsaranalysis.mcd - March 2022 pwe - Accepts text files from syn_compress.exe and plots folding analysis files for pulsar recognition

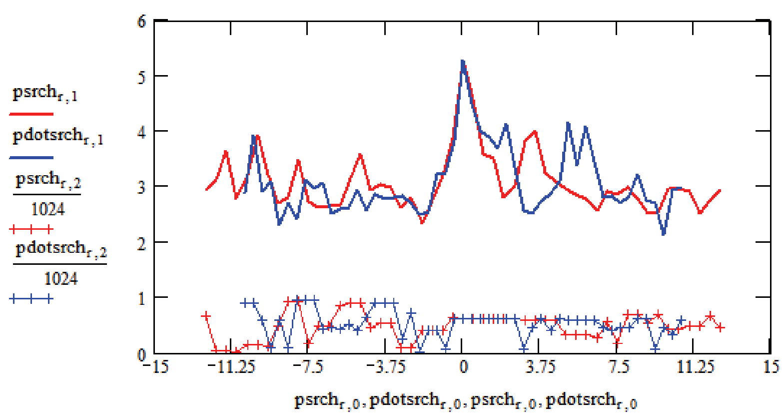
Mode = 0 Period Range Divisor = 2
Pulsar Period = 714.492092 msData clock= 2.00 ms
Pulsar Pulse Width = 6.50 ms
Pulsar Dispersion Measure = 26.70
Rolling Average Number = 8
Input file bytes =537600000
RF Centre = 611.0 MHz, RF Bandwidth = 10.0 MHz
DM Band Delay = 9.716ms Delay bins = 13.924ms
Blanked Bands: 0
Blanked Sections:
No. Data Samples = 134400000
No: FFTs = 16
Input Data per FFT channel = 8400000
No. of Output fold blocks 128 ;   bins = 1024
No. of Output samples = 131072
Working file duration = 16800 secs
Number of pulsar periods = 23513
ppm adjustment = 0.00
Max Pulse ppm drift = 0.00 ms
Compression Ratio = 183.70
Period Search Range = -12.50 ppm to 12.50 ppm
P-dot Search Range = -10.63 pp10e10 to 10.63 pp10e10

Best Profile

$k := 0..1047$



Period/P-dot Search



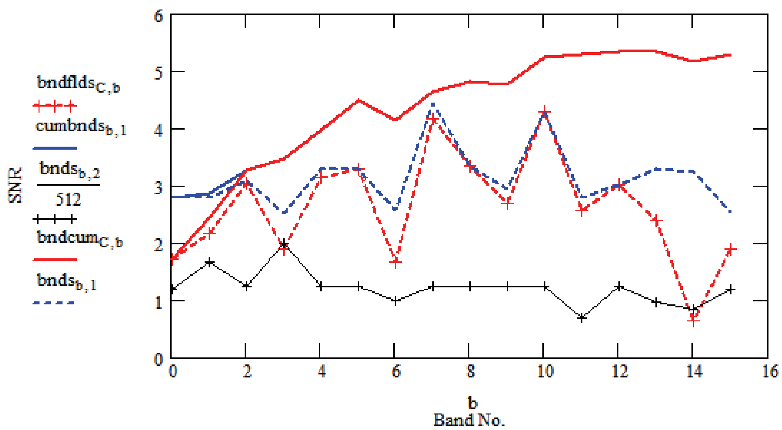
prof := READPRN("profile.txt")
psrch := READPRN("periodS.txt")
pdotsrch := READPRN("pdotS.txt")
bndflds := READPRN("bandat.txt")
cumsec := READPRN("cumsec.txt")

bnds := READPRN("bandS.txt")
dmsearch := READPRN("dmSearch.txt")
cumbnds := READPRN("cumbands.txt")
puldat := READPRN("puldat.txt")
dmsrchns := READPRN("dmSrchns.txt")

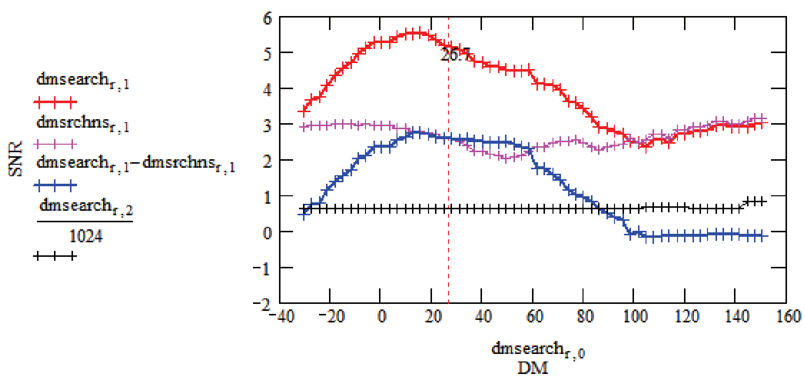
blnks := READPRN("Blanks.txt")
blnkf := READPRN("Blankf.txt")
rawdat := READPRN("rawdat.txt")

$s := 0..127$ Target Bin No. $C := 637$ $r := 0..59$ $b := 0..15$

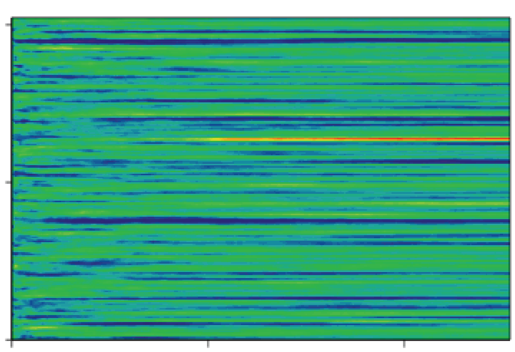
Band Properties



DM Search

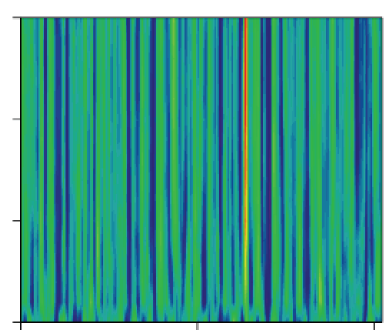


Cumulative Section SNR



foldat

Cumulative Band SNR



bndcum

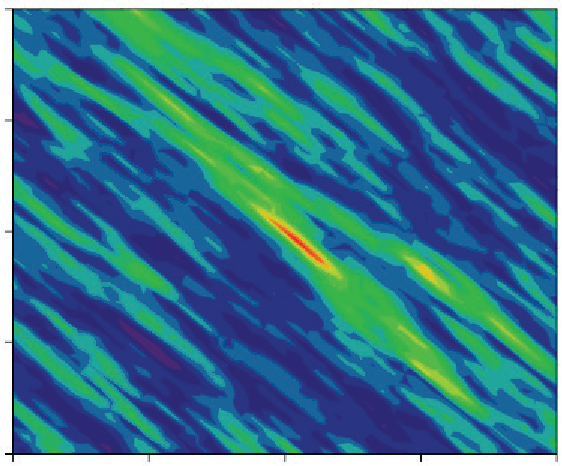
secsnr := READPRN("secsnr.txt")　　secavsnr := READPRN("secavsnr.txt")

allbands := READPRN("allbands.txt")　　bstdmprf := READPRN("dmprf.txt")

foldat := READPRN("foldat.txt")　　bndcum := READPRN("bandcum.txt")

ppd2d := READPRN("ppd2d.txt")　　rol := READPRN("secavrol.txt")

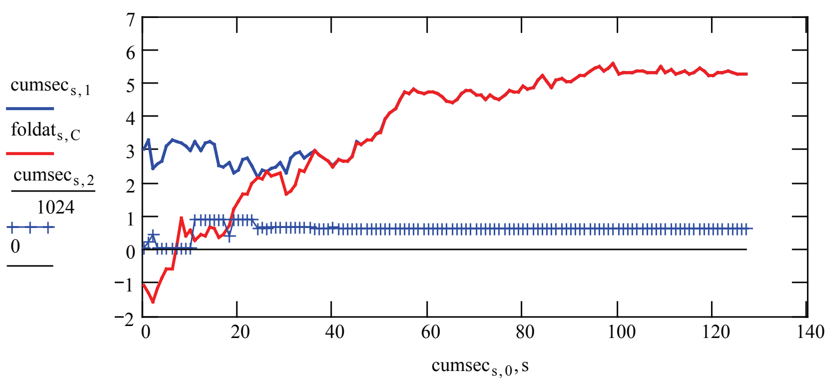
Out := READPRN("outdat.txt")　　secavfol := READPRN("secavfol.txt")



2-D Period/P-dot

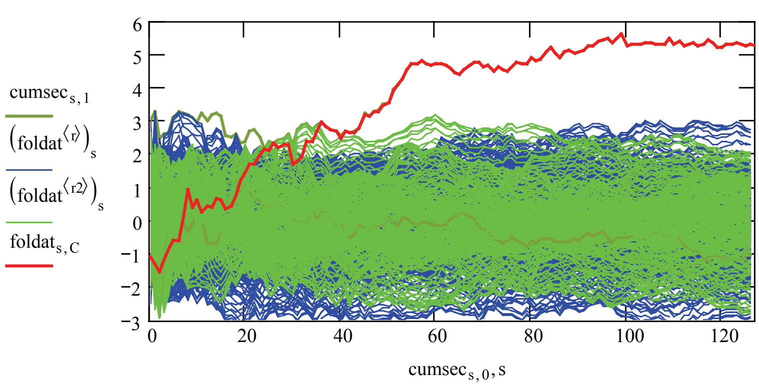
ppd2d

Cumulative Peak/Target SNR



$r := 00, 1 .. C - 20$　　$r2 := C + 20 .. 900$

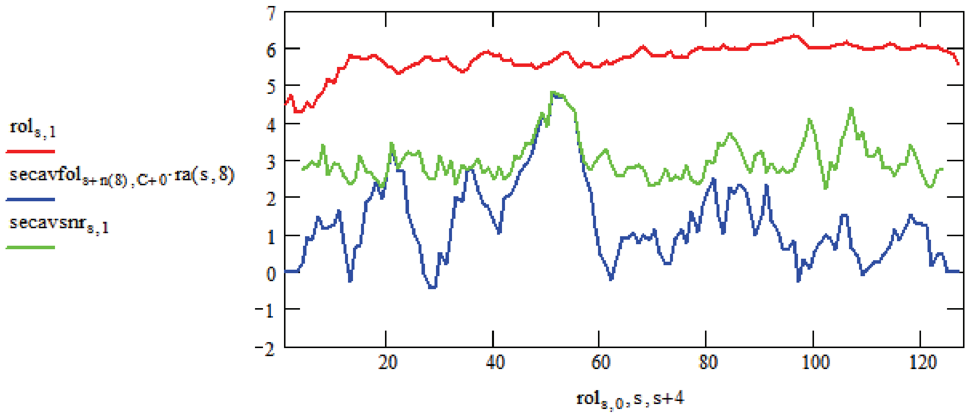
Cumulative Target and Nois



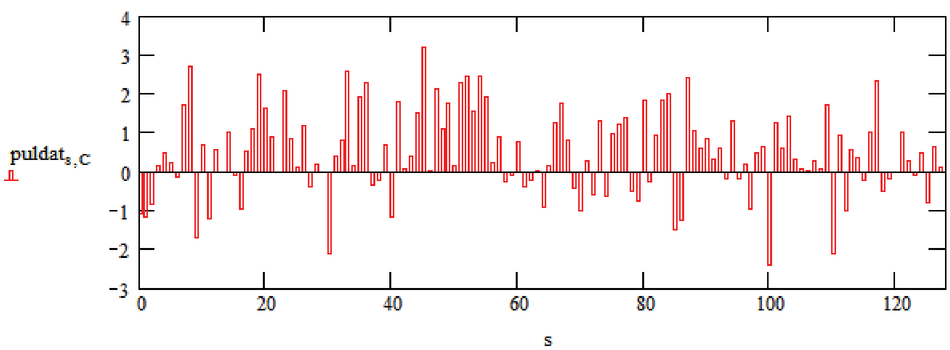
$$n(p) := -\mathrm{floor}\left[\left(\sum_{x=1}^{p-1} x - \frac{p}{2}\right) + 1\right] + 128\cdot(p-1)$$

$$ra(s,p) := \mathrm{if}\left[\left[\left(128 - \frac{p}{2}\right) + 1 > s > \frac{p}{2} - 1\right], 1, 0\right]$$

Rolling Average (RA) Search - SNR RA peak,red; section peak, green; target 8sec RA, blue



Target Section SNR



Rolling Average Responses - 4,red; 32,blue; 64,green; 96 (optimum),magenta.